NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur

Biomedical Signal Processing

Assignment- Week 7

TYPE OF QUESTIONS: MCQ/MSQ

Number of questions: 10 **Total mark: 10 X 1 = 10**

QUESTION 1:

The rectangular window is given by

a) $w_R(k)=\begin{cases}1 & \text{for } |k|\le N\\ 0 & otherwise\end{cases}$

b) $w_R(k)=\begin{cases}1 & \text{for } |k|\le N-1\\ 0 & \text{for } |k|>N\end{cases}$

c) $w_R(k)=\begin{cases}1-\frac{N}{|k|} & \text{for } |k|\le N-1\\ 0 & \text{for } |k|>N\end{cases}$

d) $w_R(k)=\begin{cases}1-\frac{|k|}{N^2} & \text{for } |k|\le N-1\\ 0 & \text{for } |k|>N\end{cases}$

Correct Answer: a

Detailed Solution: Refer slide number 25 of Periodogram

QUESTION 2:

The periodogram estimates the power at frequency f_0 by

a) Filtering data with band pass filter
b) Sampling the output
c) Computing the magnitude squared
d) None of these

Correct Answer: a, b, c

Detailed Solution: Refer slide number 13 of Periodogram

QUESTION 3:

If the impulse response of an LTI system and the input signal both are rectangular waves, then the output will be

a. Rectangular wave

b. Triangular wave

c. Ramp

d. None of the above

Correct Answer: b

Detailed Solution:

QUESTION 4:

Which of the following statement(s) is \are true when the window function has high magnitude of the side lobes in its Log-magnitude frequency response

a. Wide main-lobe width

b. Increased spectral leakage

c. Small main-lobe width

d. All of these

Correct Answer: b,c

Detailed Solution: Refer section 6.3.3 (Pg. # 363) of the book 'Biomedical Signal Analysis', by R.M. Rangayyan

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QUESTION 5:

Kurtosis is defined as

[Note: f_{m2}, f_{m3} and f_{m4} are second, third, and fourth order moments]

a. $\dfrac{f_{m3}}{(f_{m2})^{3/2}}$ b. $\left(\dfrac{f_{m2}}{f_{m2}}\right)^3$ c. $\dfrac{(f_{m3})^3}{(f_{m2})^2}$ d. $\dfrac{f_{m4}}{(f_{m2})^2}$

Correct Answer: d

Detailed Solution: Refer slide number 44 of Periodogram

QUESTION 6:

Which of the following represents correct order of waves/events occurring in carotid pulse?
(P: Percussion wave, T: Tidal wave, D: Dicrotic notch, DW: Dicrotic wave)

a. P, D, T, DW
b. P, T, DW, D
c. P, D, DW, T
d. P, T, D, DW

Correct Answer: d

Detailed Solution: Refer section 1.2.10 of the book 'Biomedical Signal Analysis', by R.M. Rangayyan

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QUESTION 7:

Which of the following statement(s) is/are true for the first order moment of a PSD

a. It is a mean frequency of a range of frequencies over which signal power is spread
b. It splits the PSD in two half's of equal energy
c. It is a useful measure of the concentration of signal power
d. All of the above

Correct Answer: a, c
Detailed Solution:

QUESTION 8:

Which of the following estimator also known as weighted covariance estimator?

a. Daniell's Spectral Estimator
b. Kalman estimator
c. Blackman-Tukey spectral estimator
d. None of these

Correct Answer: c
Detailed Solution: Refer slide number 35 of Periodogram

QUESTION 9:

Which of the following statement(s) is/are true for averaged periodogram of PSD.

a. Assumes K independent data segments
b. Mean value of the averaged periodogram is same as mean value of periodogram with any data set
c. Variance of the averaged periodogram estimator decreases with increase in K
d. Mean value of the averaged periodogram can never be same as mean value of periodogram with any data set

Correct Answer: a, b, c
Detailed Solution: Refer slide number 28 of Periodogram

QUESTION 10:

The beginning of S2 in a carotid pulse can be obtained by

a. Subtracting S2-D delay from the dicrotic notch position
b. Adding S2-D delay from the dicrotic notch position
c. By applying Pan–Tompkins algorithm to the carotid pulse
d. None of the above

Correct Answer: a

Detailed Solution: Refer section 4.10 (pg. # 285) of the book 'Biomedical Signal Analysis', by R.M. Rangayyan

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